

ULCINJ, Montenegro

WMO: 134640

Lat: 41.917N

Lon: 19.217E

Elev: 29

StdP: 100.98

Time Zone: 1.00 (EUC)

Period: 90-08

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.9	0.5	-12.3	1.3	2.6	-10.3	1.6	2.8	13.6	1.6	11.6	4.2	4.2	70	0.335

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.6	32.8	23.3	31.2	22.8	29.8	22.5	25.3	29.5	24.5	28.7	23.8	27.9	3.3	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.1	27.7	23.3	18.1	27.3	22.4	17.2	26.7	77.7	29.9	74.2	28.7	71.3	28.0	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	8.4	6.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.2	7.7	8.1	11.4	14.4	18.7	22.4	25.3	25.1	21.3	18.0	13.1	8.2		
	DBStd	6.95	3.24	3.60	3.07	2.54	2.72	3.34	2.24	2.37	2.19	3.10	3.49	3.19		
	HDD10.0	266	84	72	22	2	0	0	0	0	0	0	12	74		
	HDD18.3	1506	331	288	214	121	29	6	0	0	3	43	159	313		
	CDD10.0	2524	12	17	66	133	270	372	474	467	340	249	104	19		
	CDD18.3	722	0	0	0	2	40	128	216	209	93	33	2	0		
	CDH23.3	5205	0	0	0	3	174	959	1894	1724	374	77	1	0		
	CDH26.7	1477	0	0	0	0	27	286	600	506	52	6	0	0		
(14) Wind	WSAvg	2.4	2.8	3.2	2.6	2.2	2.1	2.0	2.0	2.0	2.2	2.5	2.5	3.4		
(15) Precipitation	PrecAvg	1403	180	145	129	124	68	55	33	51	99	152	191	175		
	PrecMax	1791	327	348	268	268	174	149	130	230	253	393	369	454		
	PrecMin	1013	1	36	28	26	10	6	0	0	8	0	62	51		
	PrecStd	235	88	73	71	61	55	43	31	55	72	108	103	95		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.1	17.5	20.2	23.5	29.1	33.8	34.6	34.3	30.1	27.2	22.6	17.8		
		MCWB	11.8	11.0	11.9	15.6	20.4	23.5	23.2	24.4	22.2	18.7	17.0	15.6		
	2%	DB	14.7	15.7	18.3	21.6	26.8	31.5	32.9	32.4	27.9	25.3	20.8	15.6		
		MCWB	11.5	10.8	12.3	14.9	18.9	23.3	22.7	23.5	20.7	19.4	16.8	12.6		
	5%	DB	13.7	14.4	17.0	20.0	25.2	29.7	31.4	30.6	26.5	23.9	19.5	14.3		
		MCWB	10.8	10.3	12.1	14.4	18.5	22.4	22.6	23.1	20.3	19.2	15.6	11.7		
	10%	DB	12.5	13.4	15.7	18.5	23.6	28.1	29.9	29.2	25.3	22.7	18.2	13.0		
		MCWB	9.5	9.9	11.8	13.9	17.9	21.8	22.3	22.6	19.9	18.4	14.8	10.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.6	13.5	14.4	17.3	21.4	26.3	26.1	26.5	23.7	22.0	19.1	16.3		
		MCDB	15.1	15.5	17.2	21.4	26.5	30.3	29.9	30.7	27.9	24.8	21.3	17.9		
	2%	WB	12.5	12.4	13.6	16.2	20.3	24.7	25.0	25.4	22.7	21.0	17.3	13.2		
		MCDB	14.1	14.5	16.2	19.6	25.2	29.2	29.1	29.6	26.2	23.5	19.8	14.9		
	5%	WB	11.2	11.5	13.1	15.4	19.5	23.5	24.1	24.7	21.7	20.0	16.2	12.2		
		MCDB	13.0	13.6	15.5	18.5	24.2	28.2	28.5	28.7	25.0	22.7	18.9	13.8		
	10%	WB	9.9	10.4	12.5	14.7	18.6	22.5	23.5	24.0	20.9	19.2	15.2	10.9		
		MCDB	12.0	12.6	15.1	17.7	22.7	27.3	28.2	27.6	24.2	22.0	17.8	12.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.6	6.9	6.4	6.8	7.3	7.5	7.6	7.3	7.2	7.0	6.6	5.6		
		MCDBR	7.8	7.8	7.9	8.8	9.7	9.7	10.3	9.1	8.8	8.1	6.9	6.5		
		MCWBR	5.6	5.7	5.0	4.7	5.2	5.1	4.7	4.8	4.9	4.8	4.7	4.9		
	5% WB	MCDBR	6.5	6.9	6.9	7.5	8.7	8.6	8.1	8.0	7.4	6.8	5.8	5.7		
		MCWBR	5.1	5.5	5.0	4.6	5.1	5.2	5.0	5.4	5.2	4.8	4.6	4.7		
(40) Clear-Sky Solar Irradiance	taub		0.341	0.364	0.406	0.456	0.444	0.425	0.426	0.440	0.422	0.394	0.367	0.342		
	taud		2.445	2.368	2.255	2.132	2.238	2.338	2.325	2.278	2.324	2.378	2.403	2.442		
	Ebn at Noon		799	833	834	816	837	852	846	822	810	788	756	761		
	Edh at Noon		78	98	122	149	137	124	125	127	113	96	81	73		
(44) All-Sky Solar Radiation	RadAvg		1.71	2.49	3.73	4.90	6.11	7.01	7.22	6.36	4.59	3.16	1.99	1.53		
	RadStd		0.18	0.37	0.49	0.49	0.45	0.42	0.33	0.35	0.34	0.30	0.19	0.26		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)		+0.38	N/A	N/A	N/A	N/A	N/A	N/A	-80	N/A	N/A

Nomenclature: See separate page